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PART III

MAKING AS INQUIRY

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Thinking as Handwork: Critical Making with Humanistic Concerns

GABBY RESCH, DAN SOUTHWICK, ISAAC RECORD,
AND MATT RATTO

Thinking Is Handwork

Late one night in the spring of 2015, two members of our lab at the University of Toronto found themselves hunched over a lab bench strewn with skeins of wire insulation and plastic shavings, brainstorming over a seemingly unsophisticated but remarkably complex challenge: How does one go about making a semi-hollow piece of plastic *feel like* a hand-worn chunk of solid ivory?¹ We experimented with a number of shop-room hacks, from buffing the object's striated surface with a mixture of animal fat and carnauba wax until it had the "worn" look and texture of aged ivory, to drilling a hole in its base and filling it with fine sand to simulate how solid ivory *weighs* in one's hand (see Figure 16.1). This exercise was part of an experiment in creating 3-D-printed replicas of ivory busts from a collection at the Art Gallery of Ontario. The busts the replicas reproduce are themselves replicas, originally made with a Victorian-era pantograph (a nineteenth-century precursor to computer numerical control [CNC] and 3-D-printing technologies) invented by Benjamin Cheverton, a notable artist, craftsperson, and engineer, for the Great Exhibition of 1851. Each replica was copied mechanically from much larger marble busts, many hand-carved by Sir Francis Chantrey, the leading portrait sculptor in Regency-era Britain.

Even if, echoing Walter Benjamin (1936), works of art have *always been reproducible*, this particular **reproduction** process stirred provocations to which **critical digital humanists** might attend. We might consider how the overwhelmingly upper-class male social network that inspired Chantrey (and, consequently, Cheverton) does not attract significant attention in an art museum filled with works of similar inspiration. Or we could interrogate the vectors along which the Victorian ivory trade, a key facet of British imperialism, produced a medium that continues to haunt museums. With a primary motivation to weigh various techniques that might make

random but this amused me because we use the same word, skein, to refer to the way yarn is wrapped + stored

resonance with Parikka (2017) and [149] Herzog + Parikka (2012) on zombie media and what is revealed ecologically, or in terms of power and extraction, when we engage in media archaeology



Figure 16.1. Gabby Resch experimenting with sand. Photograph by Dan Southwick.

our object feel more or less “real,” we had to sift through and across a **mangle of practice** (Pickering 1993) that requires a diverse array of digital pre-production techniques, exercising our own knowledge of processes—both material and digital—that constitute 3-D printing as a medium and negotiating critical issues of authenticity, curatorial and institutional authority, and the relationship between the materiality of museum artifacts and their digital-material surrogates. These concerns undergird our ongoing experiments in replication.

also
cited in
Burgess (2021)

In this chapter, we describe some ways to think productively about materiality by demonstrating how material interventions trouble theoretical work (and vice versa). We elaborate on opportunities for digital humanists as well as science and technology studies (STS) scholars who are involved (or implicated) in discussions about methods for **critical making** (Ratto 2011), especially in work that engages with cultural institutions such as galleries, libraries, archives, and museums. We describe recent experiences our lab—composed of members with backgrounds from philosophy of science and Chinese history to electrical engineering and systems administration—had building and deploying objects as well as experiences, most of which used 3-D printing as a medium, in a handful of world-class museums. These projects cut across humanities and STS themes and generally necessitate interdisciplinary tactics. Taken together, they broaden an ecology of iterative software and hardware work, interaction techniques, and scholarly practices that we regularly invoke, adapt, and reimagine. A host of fundamental issues inform this work, including how scientific and cultural knowledge are regularly co-produced in cultural institutions; how infrastructures that cross disciplinary boundaries can share objects, methods, and features along digital and material axes; and how new technologies that afford greater interaction across digital and material spaces change the ways cultural institutions engage their publics. These issues remain open and messy, but we hope our examples will contribute to a necessary dialogue in which many of us are already embedded.

Atoms, Bits, Media, Matter

When we articulate our work somewhere along a material-digital continuum, or through approaches and techniques that blur the material-digital distinction, we fall into a seductive trap. **Troubling the dichotomy between material and digital comes naturally in the academy, where we can ruminate endlessly about whether inscribed bits on a hard drive are continuous or discrete, or whether digitality is implemented over a continuous analog substrate** (Smith 2003). A significant proportion of our lab's research considers the materiality of digital infrastructures, and we generally operate from a position with which many digital humanists likely agree: that the digital is always material. Computation may be considered “a modeling activity” where correspondences between distinct domains (e.g., digital and material) must be established (Isbell et al. 198). But computation is a phenomenon where real-world effects are not just mirrored in digital space; they are entangled with digital representations. As a computational activity guided by a correspondence theory of representation—where the “closer it is to the real thing, the better” (Lynch 217)²—3-D modeling is a site where this entanglement is especially pronounced.

These dialogues are productive and enriching for scholars, but our attempts to unsettle the marked distinction between digital and material culture meet resistance from cultural heritage professionals who have to reconcile this distinction in their

work. For many of these professionals, “the digital” is necessary but also threatening. Digital culture can challenge their institutional mandates (e.g., “Why should the public go to a museum when they can see the same content on an iPad at home?” is a familiar refrain, however shortsighted it might be). Rather than thinking of material and digital as distinct entities that meet, overlap, and occasionally blur into each other, we encounter them as entangled phenomena, situated in a gradient along which agential possibilities are distributed among human and non-human entities and cut by a wide and ever-changing array of social, technical, and institutional concerns (Barad 2003). Compounding this digital-material entanglement, we also think of 3-D reproduction as a unique medium that actively figures and influences the processes through which digital objects are reshaped into analog or material ones.

These factors influence how we make critically, especially in humanities contexts. We engage, often unequally, with ideas from STS, critical theory, digital humanities (DH), design, feminist technoscience, human-computer interaction (HCI), maker cultures, and a number of cognate disciplines and movements. As our projects indicate, critical making invokes a reflexive methodological position that interrogates the politics of sociotechnical objects and systems as well as the institutions that we engage and care about—particularly as those institutions are forced to navigate a rupture in how the public perceives their relevance at a time of increasing austerity. In the sense that we draw selectively upon specific methods, requiring hands-on engagement as an additional resource, theory both guides and emerges from the critical making projects our lab undertakes.

This making does not have to be “digital” or even **participatory making** (as exemplified by the so-called “maker movement”), but it should attempt to reconcile a schism between those who purportedly create digital (and digital-derived) content in institutional contexts and those purportedly tasked with representing and interpreting it (as if these tasks are discrete). For us, “doing theory” with critical making³ entails moving beyond shallow critical reflection, or mere acknowledgment of the relational ontologies brought to bear on the institutional contexts of our work. ^{→ + what is produced when we take that move beyond? What are the explicit goals of that move?} However, theory cannot be taken for granted. Many reports seem to give passing reference to theoretical perspectives, rather than demonstrating deep engagement with them. This tendency is especially problematic in projects that illuminate unequal power distributions and then claim to somehow grant greater agential capacity to actors who were previously rendered invisible, mute, or obsolete (as if agency is something we, as makers or scholars, can grant⁴). To avoid this tendency, our projects use theoretical concepts in occasionally utilitarian, but generally non-deterministic, ways to think through the agential possibilities, temporal conditions, and materiality of institutions.

While we do not intend for our critical making methods to be grounded in instrumental logic, we constantly negotiate between instrumental and aesthetic concerns when working with cultural institutions. Considerations abound, including how we can *use* 3-D printing to make museum objects “more interactive.”

Traditional value propositions and epistemic commitments we might encounter in scientific and artistic communities—utility and empirical rigidity, on the one hand; expression and empirical flexibility, on the other—are troubled by considerations toward assessment. We are also not strictly beholden to questions of aesthetics, the pragmatics of developing institutional relationships, or arguments about the relative value of processual acts versus evocative objects (although we lean toward the former). Among the most interesting discoveries we have encountered in our engagements with cultural institutions is a heightened sensitivity to how preparing an object for display can be a more important locus of epistemic activity than the exhibited object itself.

We are not going to pretend to have a definition of DH that works for everyone, nor are we as intimately familiar with DH concerns as many readers of this book might be. If we understand critical computational inquiry in the humanities as both the interrogation of digital tools used by humanities scholars and the use of digital tools to interrogate theoretical claims about how cultural reproduction occurs, then we might assume that creating and understanding digital archives of cultural “data” should be a core concern of the H/DH archipelago (given that some of the most mature and robust computational tools reside in this space). What are the intersections of “humanistic concerns and digital capabilities” (Drucker, *Spec-Lab* 64) where critical making might present valuable insight? Should there be an agreed-upon description of “humanistic concerns” that we can take as a starting point? What might some “acceptable” maker activities for humanities scholars be if such activities use immature technologies that have not received official institutional sanction? Is humanities scholarship better served by taking *the act of* prototyping as a form of pedagogy, or by insisting on the release of full-fledged investigative tools into the wild, where so many of them meet a lonely death? Generally speaking, we have not reconciled our work with claims that tools, if they are to be taken as theories, should be accessible, transparent, and available to peer review, although we tend to agree with such claims (Ramsay and Rockwell 2012). For the most part, we are also sensitive to a critique that using computers to manipulate or interpret humanities data does not always constitute theorizing.

With these points in mind, we propose that 3-D rematerialization (including additive manufacturing such as 3-D printing as well as subtractive manufacturing such as routing and laser cutting) is a kind of visualization technique appropriate for certain humanities concerns. We undertook a handful of recent projects guided by this consideration. One project involved a collaboration between members of our lab and an international team of researchers and artists at the Art Gallery of Burlington. We captured embodied gestural data performed during craft practice and artistically reproduced it through a series of 3-D prints. Another project involved members of our lab collaborating with the Royal Ontario Museum to produce 3-D–printed tactile representations of 2-D digital images that have been included in an annual photography exhibit and are now in the museum’s teaching collection. This project,

which began by algorithmically displacing and manipulating color values in digital images but ultimately required a significant amount of hand “sculpting” that traversed the material-digital gap, introduced a new mode of sensory engagement for non-sighted (as well as sighted) visitors to the museum. Despite the success of these collaborations, we are not particularly “eager to suspend critical judgment in a rush to visualization” (Drucker, “Humanities Approaches to Graphical Display” 1), likely because prototypes and half-finished iterations generally satisfy the requirements of our work. Beyond the issues listed earlier, a host of possibilities, affordances, and opportunities inherent in 3-D work require greater attention: how, for instance, 3-D–driven investigations inform recent dialogue around “making” in humanities scholarship; what 3-D–based curricula across a full chain of activities—from capture (digitization) to cleanup and design (digital artisanship) to printing (rematerialization) to finishing (material artisanship)—might look like for museum and DH professionals; and how scholars should engage the public in participatory and collaborative 3-D making experiences that promote sustained engagement with cultural materials.

The 3-D Museum Medium

The fact that many museums are already well versed in 3-D scanning and printing technologies (for conservation purposes, at least), and that museums are *already* 3-D in the first place, should temper the hype about 3-D printing as an “ideal” technology for museum-based humanistic inquiry. But there is an interesting—albeit troubling—trend under way in which 3-D printing is positioned as a technology that museums might (and, in some cases, should) embrace to develop new modes of interaction between humans and artifacts (Sportun 2014). While 3-D printing has inhabited a role as one of the preeminent maker technologies, its integration in cultural spaces necessitates a host of critical considerations, including questions about the materiality of the medium (e.g., what happens to all this plastic ephemera, particularly when printing happens on-site, as it off-gases or disperses in the presence of priceless artifacts?), the expertise of operators (e.g., is 3-D printing a black box-able technology, as numerous manufacturers hope, or does it require the kind of tacit knowledge, experience, and care that comes from opening its hardware?), and who it ultimately serves (e.g., are STEM-funding agencies herding museums into initiatives that promote training members of the public to meet the world’s crises, and, if so, who in the museum benefits from these neoliberal narratives?). While there is a growing body of creative and playful examples of museum-based 3-D, we can offer scant evidence of the medium’s use in critical provocations that challenge the institution itself.

considerations for
affordances +
limitations of
employing 3D printing
technologies as part of
cultural heritage work

Both the museum world and the public are witnessing a paradigm shift in how museums are perceived (Anderson 2004)—from collections-centered, traditional institutions to visitor-centered, reimagined public institutions. The two paradigms

are not mutually exclusive, and some recent evidence suggests that museums might cautiously warm to participatory culture without disregarding their past mandates (Söderqvist 2010). This idea is contested, with evidence of backlash toward the participatory museum.^(Such as? and by whom?) Concomitant with this paradigm shift are challenges to not only curatorial and institutional authority but also the power of objects to tell stories and the very definition of an exhibit. As critical STS scholars with humanistic leanings, we are naturally drawn to projects that illuminate the politics of institutional authority. **So how do we reconcile our critical work with the politics of our institutions?** Consider the following example.

In 2013, members of our lab collaborated with a team of partners to create and implement a 3-D design and printing experience at the Royal Ontario Museum (ROM). Visitors to the museum were invited to use iPads to create unique pieces of an imaginary Mesopotamian city that were 3-D-printed *in situ* and installed in a dynamic, collective, participatory exhibit. A number of unanticipated situations emerged throughout the run of the exhibit, which was staged on Friday nights as part of a public engagement that turns the museum into a kind of nightclub for thousands of (mostly young) adults. A significant number of participants experimented with deliberately anachronistic designs, many resembling modern skyscrapers. Others collapsed columns into ziggurats or turned temples upside down. One visitor referred to a sleek, modernist obelisk they created as a “Mesopotamian Freedom Tower.” This particular example would probably horrify curators or museum interpreters, who are not only accountable to visitor experiences but also responsible for representations of history and culture. We, however, were presented with an opportunity to think critically about how museums use new technologies to structure temporality and present authoritative institutional narratives and, furthermore, to consider whether these narratives are commensurable with claims that technologies like 3-D printing afford the ability to re-create, or even reimagine, the past.

In another example, described in the opening paragraph of this chapter, we pre-framed^{— as well as drawn on those understandings of the past to future imaginaries} touchable models for an exhibit at the Art Gallery of Ontario (AGO). This exhibit, which highlights busts of notable figures such as James Watt (a contemporary of Cheverton), might initially be read through a kind of *Leviathan and the Air-Pump* (1985) **network analysis**, where scholars look for evidence of **epistemic cultures** in the material record of the collection. Historiographical work that intersects with critiques of contemporary **curatorial regimes** is also interesting in this instance. What and whose histories are collected and represented, and how are they troubled when scholars and the public reimagine their mode of interaction with exhibits? The experience of preparing pieces for an exhibit that speaks to historical power dynamics and modes of production, as well as contemporary meta-questions about the role of technology in museum interactions, was illuminating for us. It informed ongoing theoretical discussions members of our lab have about histories of **ocularcentric** interaction and the politics of touch in **memory institutions**. Additionally, it forced us, as STS scholars with critical humanistic tendencies,

possible connection to Muñoz's (2009) queer futurity?

to look inward, at our own practices, relationships, and epistemic commitments. In addition to its use in participatory culture, 3-D printing is increasingly proposed as a technology that affords new modes of interaction that then increase inclusivity or accessibility. For instance, the AGO exhibit included a projected video of our 3-D making process, positioned adjacent to a touchable bust. The video performed a mediating role that called attention to the gallery's willingness to use new technologies for the purposes of accessibility and perceived relevance. A similar video was displayed alongside the tactile models we prepared for the ROM. Museums are beginning to foreground these aspects of 3-D printing in their marketing, suggesting that 3-D printing technology somehow connects them with the past and the future at the same time, while also extending their collections to a wider audience. These uncomplicated narratives can seem scripted or determined by the manufacturers of maker technologies, and we are not entirely comfortable being associated with them.

Returning to the problem of conceptualizing a material-digital entanglement that is attentive to institutional politics, we recognize a growing tendency toward characterizing 3-D-printed objects as digital instructions implemented in a **material substrate**. This tendency overlooks and often erases the considerable amount of **handwork** that goes into the digital composition and preparation of printed artifacts. Such handwork includes the embodied act of capturing an object and digitizing it, the somewhat **skeuomorphic** enactment of a digital drafting station that requires a pressure-sensitive stylus and tablet to digitally sculpt and prepare scanned data, and post-processing work that entails cleaning up random bits of digital noise that get materialized (something the AGO also highlighted in their video display, as their videographer focused on one of our hands trimming residual bits of plastic). But this labor is not just handwork. It requires a more sophisticated temporal understanding of what it means to do material work. The *matter* of 3-D printing resists stability—it desires to disperse, settle, and blend, generating latent chemical effects that humans must be wary of, particularly conservationists who have to weigh how 3-D-printed objects might off-gas, or how the process, if done *in situ*, might emit ultrafine particles into a museum space.⁵ Offering assurances that the wild west of 3-D printing is supported by a few decades of anecdotal evidence is insufficient. We must account for why 3-D printing would be an institutional issue in the first place.

Making Things Matter Together

We take Alan Liu's (2012) suggestion that DH can learn from how STS approaches culture, technology, and society. Liu proposes that engaging with STS "would help the digital humanities develop an understanding of instrumentalism—including that of its own methods—as a culture embedded in wider culture" (502). Further, we argue that critical making with humanistic concerns could serve as a bulwark against a tendency by scholars, humanists among them, to replace **technological determinism** with **social determinism** (Latour 84). Liu writes, "Only by creating a

methodological infrastructure in which culturally aware technology complements technologically aware cultural criticism can the digital humanities more effectively serve humanists by augmenting their ability to engage today's global-scale cultural issues" (502). DH and STS can do work together that might enable us to productively intervene in society. We could devise joint tactics to face the "problem" of digitization. We could expose flawed, deterministic tendencies that pervade our disciplines. We could highlight the troubling conflation of digital literacy with critical socio-technical and information literacy, when these mean quite different things but are often used interchangeably.

A plethora of worthwhile scholarship exists under the aegis of DH, STS, and even HCI that uses "making" as a productive and reflexive process. Together, we might extend the reflexive modes of engagement that are boundary-agnostic. Surely, some of the "little hacks" we deploy in STS-grounded work can be ported to humanistic contexts. A far more interdisciplinary framing than DH-meets-STS is required, however, if we are going to construct the methodological apparatus that Liu proposes. (This framing would include recognizing projects in adjacent fields like HCI that have pursued similar goals.)⁶ For our lab, such framing draws from contemporary museology, interaction design, media theory, and STS's increasingly distant parent, philosophy of science, which, as Willard McCarty notes, hovers in the background of DH (17).

Following Karen Barad, we recognize an "entanglement of matter and meaning" that questions erroneous separations of nature from culture as well as matters of fact from matters of concern (Latour 2005) and care (Puig de la Bellacasa 2011).⁷ This entanglement is especially important if matters of fact—nature, raw data, positivist empirical study, and scientific realism, for instance—can no longer be restricted to the natural sciences, and matters of concern—culture, social constructivism, and values, for example—have moved beyond the humanities and social sciences. Like Barad, we seek to recognize how the humanities and sciences are already entangled, and to trace diffraction patterns that record the histories of interaction, interference, reinforcement, and difference between them. This approach requires taking into account how each "culture" recognizes another significant entanglement—between *matter* and *relevance*. Do the concerns that matter to us, as humanists and humanities allies, conflict with agendas that define how our work is deemed relevant? We must ask to whom we want to be accountable, as well as how the epistemic products of our work produce a kind of relevance that can be tough to measure. In the interest of staging a "productive intervention in society," we must also ask: What kind of advocacy work do we want to do? Maybe we need to develop pervasive technologies and experiences that, as Liu suggests, "fundamentally reimagine humanities advocacy" (497) and enhance the ability of humanistic scholarship to both communicate and resonate with the general public. In that case, sites such as museums, where the public effectively meets the humanities face-to-face, would be productive sites of intervention.

We must also unbox the **epistemic objects** (Knorr Cetina 2008) that humanistic and STS-informed making might share, or the **boundary objects** (Star and Griesemer 1989) that might enable translation across epistemic disjunctures. Critical making, as we propose it, is more a mode of critically reflexive (even transformative) inquiry than a record of scholarly activity. While tracing scholarly knowledge production might be a core concern of both STS and DH, our work is not directly related to it. Writing object biographies of digital making experiences may be a “way of knowing,” and situating researchers in these unwieldy biographies may be a political act (Morgan 2012), but we also need to look for theories in prototypes and, following Galey and Ruecker (2010), recognize that our projects may be interpreted as rhetorical devices to “be used in persuasive performances” (Ramsay and Rockwell 78).

Critical making examines objects that are never fixed. Direct material engagement is a “condition for knowing,” and this knowing is not only “knowing from a distance” (Dolphijn and van der Tuin 52). It is a combination of **distant and close reading**, attending to the particulars of institutional artifacts and the networks through which they travel. It provokes thoughtful studies of murky suspensions, where bits of material and digital worlds collide. Moreover, it encourages an examination of the different value propositions associated with reflexive material and semiotic engagement. The projects we have highlighted force us to revisit many of the Benjaminian notions we might entertain about digital reproduction and simulation, while also offering new insights into concerns, such as the increasingly algorithmic nature of DH work, that should interest humanities scholars. Is, for instance, the handwork of 3-D always an alternative to the algorithmic when much of it is done procedurally through computer-aided design?

If STS and DH scholars are going to successfully collaborate using modes of engagement similar to the ones we have described, then we must first examine our own goals. Are we looking to address critical questions or demonstrate them? Are we illustrating how things come to matter, or are we exploring how they come to matter through process? Understanding a critical issue does not automatically count as an intervention, just as a critical intervention does not automatically ensure understanding. We hope that, by mandating direct material engagement as a condition for knowing, a critical making approach to humanistic concerns can do away with some of these unnecessary confluences. The projects mentioned here offer new avenues into the politics of representation, exhibition, memory, and institutional knowledge. At the same time, they serve as platforms for critically interrogating the sociotechnical systems that are increasingly foisted on institutions struggling to demonstrate their relevance.

We need to be mindful of the multiple meanings of *audience* in these endeavors, though. If the benefits to be derived are often for the makers and, potentially, their scholarly audience, then does critical making count as scholarship, pedagogy, advocacy, or all of the above? It is often hard to tell. We can be sure, however, that this

is a perfect moment to find oneself hunched over a messy lab bench at two in the morning, mesmerized by the cacophony of Cartesian robots dancing back and forth, and trying to meet some institution's deadline while pondering the entanglements of meaning and matter, subject and object, material and digital, and—of course—handwork and headwork.

NOTES

1. This heading and the title of this chapter are inspired by two things: a passage on the “thingly character” inherent in Heidegger's assertion that “denken ist/als handwerk” (Simon 188–89) and an Instagram post by Natalie Jeremijenko: <https://instagram.com/p/4xDWUC7h>.

2. And, if so, then how do we reconcile representational theories and 3-D modeling with Johanna Drucker's assertion that “rendering observation (the act of creating a statistical, empirical, or subjective account or image) as if it were the same as the phenomena observed collapses the critical distance between the phenomenal world and its interpretation, undoing the basis of interpretation on which humanistic knowledge production is based. We know this. But we seem ready and eager to suspend critical judgment in a rush to visualization”? (“Humanities Approaches to Graphical Display” 1).

3. See Matthew Kirschenbaum's tweet from January 4, 2013 on making as doing theory: <https://twitter.com/mkirschenbaum/status/287280574815666177>.

4. See Barad on reworking the notion of agency in ways that are appropriate to relational ontologies, recognizing agency as a matter of possibilities for reconfiguring entanglements: “The notion that there are agents who have agency, or who grant agency, say, to non-humans (the granting of agency is an ironic notion, no?), pulls us back into the same old humanist orbits over and over again. And it is not easy to resist the gravitational force of humanism, especially when it comes to the question of ‘agency.’ But agency for me is not something that someone or something has to varying degrees” (in Dolphijn and van der Tuin 54).

5. We might consider this phenomenon with regard to Jane Bennett's articulation of a vibrant materiality: “as much force as entity, as much energy as matter, as much intensity as extension” (20).

6. See, for example, Sengers, et al.

7. See Barad in Dolphijn and van der Tuin (50).

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